1 Motivation

In the current 2D cylindrical model, the radiation temperature field inside the foam is decomposed into an axial Marshak wave solution and a radial eigenfunction. The radial profile satisfies a diffusion eigenvalue problem with an effective boundary condition at the foam–wall interface $r = R$.

The existing approach computes the wall albedo α_w from the wall energy loss parameterization and feeds it into the eigenvalue equation as a parameter. However, α_w itself depends on the interface temperature T_{int}, which in turn depends on the radial profile and hence on α_w—creating a circular dependency.

Here we propose a **self-consistent** formulation: apply a proper Marshak boundary condition at the radial interface $r = R$ using the known temperature profile from $r = 0$ to $r = R$. This yields a Robin boundary condition that simultaneously determines the radial eigenvalue κ_0, the interface temperature T_{int}, and the albedo α_w.

2 Setup: Radial Temperature Profile

2.1 Radiation diffusion in cylindrical geometry

Consider a cylindrical foam of radius R, with a radiation temperature field $T(r, z, t)$. At a fixed axial position z and time t, the radiation energy density in the diffusion approximation satisfies:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\, D(r)\, \frac{\partial U}{\partial r}\right) = S(r) \tag{1}$$

where $U = aT^4$ is the radiation energy density and $D = c/(3\kappa_R)$ is the diffusion coefficient with $\kappa_R = 1/\lambda_R$ the Rosseland mean opacity.

2.2 Separation of variables

We separate the radial dependence by writing:

$$U(r, z, t) = U_0(z, t) \cdot \Phi(r) \tag{2}$$

where $U_0(z, t)$ is the on-axis ($r = 0$) radiation energy density. The radial eigenfunction $\Phi(r)$ satisfies:

$$\frac{1}{r}\frac{d}{dr}\left(r\, \frac{d\Phi}{dr}\right) + \kappa_0^2\, \Phi = 0 \tag{3}$$

with $\Phi(0) = 1$ (normalization) and $\Phi'(0) = 0$ (symmetry). The general solution regular at $r = 0$ is:

$$\boxed{\Phi(r) = J_0(\kappa_0\, r)} \tag{4}$$

where J_0 is the Bessel function of the first kind and κ_0 is the radial eigenvalue, to be determined by the boundary condition at $r = R$.

3 The Radial Marshak Boundary Condition

3.1 Half-range fluxes at the interface

At the foam–wall interface $r = R$, we decompose the radiation field into outgoing (foam $\to$ wall) and incoming (wall $\to$ foam) half-range fluxes. In the P_1 (diffusion) approximation:

$$J^+(R) = \frac{c}{4}\,U(R) + \frac{1}{2}\,F_{\rm net}(R) \qquad \text{(outgoing: foam} \to \text{wall)} \tag{5}$$

$$J^-(R) = \frac{c}{4}\,U(R) - \frac{1}{2}\,F_{\rm net}(R) \qquad \text{(incoming: wall} \to \text{foam)} \tag{6}$$

where the net diffusive flux (positive outward) at the surface is:

$$F_{\rm net}(R) = -D\,\frac{\partial U}{\partial r}\bigg|_{r=R} = -\frac{c}{3\kappa_R}\,\frac{\partial U}{\partial r}\bigg|_{r=R} \tag{7}$$

Since T decreases radially ($\partial U/\partial r < 0$), we have $F_{\rm net}(R) > 0$: net energy flows outward into the wall.

3.2 Wall reflection condition

The wall absorbs a fraction $(1 - \alpha_w)$ of the incident flux and reflects (re-emits) a fraction α_w:

$$J^-(R) = \alpha_w\,J^+(R) \tag{8}$$

3.3 Combining the conditions

Substituting Eqs. (??)–(??) into (??):

$$\frac{c}{4}\,U(R) - \frac{1}{2}\,F_{\rm net}(R) \;=\; \alpha_w\left[\frac{c}{4}\,U(R) + \frac{1}{2}\,F_{\rm net}(R)\right] \tag{9}$$

Rearranging:

$$\frac{c}{4}\,U(R)\,(1 - \alpha_w) \;=\; \frac{1}{2}\,F_{\rm net}(R)\,(1 + \alpha_w) \tag{10}$$

$$\boxed{U(R) = \frac{2(1 + \alpha_w)}{c\,(1 - \alpha_w)}\;F_{\rm net}(R)} \tag{11}$$

Substituting $F_{\rm net} = -\frac{c}{3\kappa_R}\frac{\partial U}{\partial r}\big|_R$:

$$U(R) = -\frac{2(1 + \alpha_w)}{3\kappa_R\,(1 - \alpha_w)}\;\frac{\partial U}{\partial r}\bigg|_{r=R} \tag{12}$$

This is a **Robin (mixed) boundary condition**. Defining the *extrapolation length*:

$$\boxed{\ell_{\rm ext} = \frac{2(1 + \alpha_w)}{3\kappa_R\,(1 - \alpha_w)}} \tag{13}$$

the boundary condition becomes:

$$\Phi(R) = -\ell_{\rm ext}\;\Phi'(R) \tag{14}$$

4 Modified Eigenvalue Equation

4.1 Substituting the Bessel solution

With $\Phi(r) = J_0(\kappa_0 r)$ and $\Phi'(r) = -\kappa_0 J_1(\kappa_0 r)$, the Robin BC (**??**) becomes:

$$J_0(\kappa_0 R) = \ell_{\mathrm{ext}}\, \kappa_0\, J_1(\kappa_0 R) \tag{15}$$

Substituting ℓ_{ext} from (**??**):

$$\boxed{J_0(\kappa_0 R) = \frac{2(1+\alpha_w)}{3\kappa_R\,(1-\alpha_w)}\;\kappa_0\; J_1(\kappa_0 R)} \tag{16}$$

Defining the dimensionless variable $x = \kappa_0 R$:

$$J_0(x) = \frac{2(1+\alpha_w)}{3\kappa_R R\,(1-\alpha_w)}\; x\; J_1(x) \tag{17}$$

4.2 Connection to the existing eigenvalue equation

The existing code in `eigen_bessel_solver.py` solves:

$$x\, J_1(x) = \varepsilon\, J_0(x) \tag{18}$$

with:

$$\varepsilon = \frac{3}{4}\,(1-\alpha_w)\,\frac{R}{\lambda_R} = \frac{3}{4}\,(1-\alpha_w)\,\kappa_R\, R \tag{19}$$

Rewriting (**??**):

$$x\, J_1(x) = \frac{3\kappa_R R\,(1-\alpha_w)}{2(1+\alpha_w)}\; J_0(x) \tag{20}$$

So the eigenvalue equation is equivalent to (**??**) with ε *redefined* as:

$$\boxed{\varepsilon_{\mathrm{new}} = \frac{3\kappa_R R\,(1-\alpha_w)}{2(1+\alpha_w)} = \frac{3}{2}\,\frac{1-\alpha_w}{1+\alpha_w}\,\frac{R}{\lambda_R}} \tag{21}$$

Comparison with the current formula:

$$\varepsilon_{\mathrm{current}} = \frac{3}{4}\,(1-\alpha_w)\,\frac{R}{\lambda_R} \tag{22}$$

$$\varepsilon_{\mathrm{new}} = \frac{3}{2}\,\frac{1-\alpha_w}{1+\alpha_w}\,\frac{R}{\lambda_R} \tag{23}$$

For high albedo ($\alpha_w \to 1$), both $\varepsilon \to 0$, giving $\kappa_0 \to 0$ and a flat radial profile (no lateral losses). For $\alpha_w = 0$ (perfect absorber):

$$\varepsilon_{\mathrm{current}}\Big|_{\alpha_w=0} = \frac{3}{4}\,\frac{R}{\lambda_R} \qquad \text{vs.} \qquad \varepsilon_{\mathrm{new}}\Big|_{\alpha_w=0} = \frac{3}{2}\,\frac{R}{\lambda_R} \tag{24}$$

The new formula gives $2\times$ larger ε for $\alpha_w = 0$, which is the correct Marshak extrapolation length $\ell_{\mathrm{ext}} = 2/(3\kappa_R)$.

5 Limiting Behaviour

- **Perfect reflector ($\alpha_w = 1$):** $\ell_{\rm ext} \to \infty$, so $\varepsilon \to 0$. The eigenvalue $\kappa_0 \to 0$, giving $\Phi(r) = J_0(0) = 1$ everywhere—a flat radial profile. The wall reflects all incident radiation, so there is no radial loss. ✓
- **Perfect absorber ($\alpha_w = 0$):** $\ell_{\rm ext} = 2/(3\kappa_R)$, recovering the standard Marshak extrapolation length for a vacuum boundary.
- **Optically thick foam ($\kappa_R R \gg 1$):** $\varepsilon \gg 1$ and $\kappa_0 R \to j_{0,1} \approx 2.4048$ (the first zero of J_0), recovering the Dirichlet condition $\Phi(R) = 0$.
- **Optically thin foam ($\kappa_R R \ll 1$):** $\varepsilon \ll 1$ and $\kappa_0 R \ll 1$, giving an approximately flat profile with a small correction $\Phi(r) \approx 1 - \kappa_0^2 r^2/4$.

6 Self-Consistent Interface Temperature

6.1 The key idea

Once the eigenvalue κ_0 is determined from (**??**), the interface temperature follows directly from the radial profile:

$$\boxed{T_{\rm int}^4 = T_{\rm axis}^4 \cdot J_0(\kappa_0 R)} \tag{25}$$

where $T_{\rm axis} = T_{\rm axis}(z,t)$ is the on-axis temperature from the 1D Marshak march.

6.2 Self-consistent albedo from the radial flux

The net outward radiation flux at the interface is:

$$\begin{aligned} F_{\rm net}(R) &= -\frac{c}{3\kappa_R}\,\frac{\partial(aT^4)}{\partial r}\bigg|_{r=R} \\ &= \frac{c\,a}{3\kappa_R}\,\kappa_0\,T_{\rm axis}^4\,J_1(\kappa_0 R) \end{aligned} \tag{26}$$

The power absorbed by the wall per unit axial length at position z is:

$$\dot{E}_w(z) = 2\pi R\,(1-\alpha_w)\,J^+(R) = 2\pi R\left[\frac{c}{4}\,a\,T_{\rm int}^4 + \frac{1}{2}\,F_{\rm net}(R)\right](1-\alpha_w) \tag{27}$$

Alternatively, one can compute the absorbed flux purely from the foam-side quantities, without needing the wall loss parameterization:

$$\dot{E}_w = (1-\alpha_w)\,J^+(R) = J^+(R) - J^-(R) = F_{\rm net}(R) \tag{28}$$

This is a key result: the net radial flux at the interface equals the wall energy absorption rate per unit area. Since we can compute $F_{\rm net}(R)$ from the known Bessel profile, we can bypass the wall loss parameterization $E_w(T_0,t)$ entirely for the albedo calculation.

6.3 Self-consistency loop

The procedure is:

1. **Start** with an initial guess for α_w (e.g., from the previous timestep, or from the wall loss parameterization).

2. **Compute** ε_{new} from Eq. (**??**) using the current α_w, κ_R, and R.

3. **Solve** the eigenvalue equation (**??**) for κ_0 (already implemented in `eigen_bessel_solver.py`).

4. **Extract** the interface temperature from Eq. (**??**): $T_{\text{int}} = T_{\text{axis}}\,[J_0(\kappa_0 R)]^{1/4}$.

5. **Compute** the wall energy loss $\dot{E}_w$ using the wall loss parameterization evaluated at T_{int}:
$$\dot{E}_w(z,t) = \left.\frac{dE_w}{dt}\right|_{T_0=T_{\text{int}},\, t_{\text{exp}}} \tag{29}$$
(e.g., $E_w^{\text{Au}} = 0.59\,T_0^{3.35}\,t_{\text{ns}}^{0.59}$ differentiated with respect to t).

6. **Update** the albedo using the Rosen formula:
$$\alpha_w^{\text{new}} = 1 - \frac{\dot{E}_w}{\sigma_{\text{SB}}\,T_{\text{int}}^4} \tag{30}$$

7. **Iterate** steps 2–6 until $|\alpha_w^{\text{new}} - \alpha_w^{\text{old}}| < \delta$ (typically 2–3 iterations suffice).

7 Physical Interpretation

7.1 Why this is better

The current approach has a conceptual issue: the albedo α_w is computed from the 1D surface temperature T_s (the Marshak march solution), but this temperature corresponds to the *on-axis* value at $r = 0$. The actual temperature at the foam–wall interface is lower due to the radial profile:
$$T_{\text{int}} = T_{\text{axis}} \cdot [J_0(\kappa_0 R)]^{1/4} < T_{\text{axis}} \tag{31}$$

Since the wall loss scales as a high power of temperature ($\dot{E}_w \propto T^{3.35}$ for gold), using T_{axis} instead of T_{int} **overestimates** the wall energy loss, which in turn **underestimates** the albedo.

The self-consistent approach corrects this by:

1. Using the physically correct interface temperature.

2. Coupling the albedo and radial profile consistently through the Robin BC.

3. Naturally recovering the correct limiting behaviours (perfect reflector/absorber, optically thick/thin).

7.2 Geometric picture: extrapolated endpoint

The Robin BC (**??**) is equivalent to saying that the eigenfunction $\Phi(r) = J_0(\kappa_0 r)$ extrapolates to zero not at $r = R$ but at an *effective radius*:

$$R_{\text{eff}} = R + \ell_{\text{ext}} = R + \frac{2(1+\alpha_w)}{3\kappa_R(1-\alpha_w)} \tag{32}$$

- For $\alpha_w = 0$: $R_{\text{eff}} = R + \frac{2}{3\kappa_R}$ (standard diffusion extrapolation length).
- For $\alpha_w \to 1$: $R_{\text{eff}} \to \infty$ (flat profile, no effective boundary).

This is the cylindrical analogue of the classical "extrapolated endpoint" in planar slab diffusion theory.

8 Summary of Key Equations

1. **Radial profile:** $T^4(r, z, t) = T^4_{\text{axis}}(z, t) \cdot J_0(\kappa_0 r)$
2. **Eigenvalue equation (unchanged form):**
$$x\, J_1(x) = \varepsilon\, J_0(x), \qquad x = \kappa_0 R$$
3. **Modified ε from the radial Marshak BC:**
$$\varepsilon = \frac{3}{2}\,\frac{1-\alpha_w}{1+\alpha_w}\,\frac{R}{\lambda_R}$$
4. **Interface temperature:**
$$T_{\text{int}} = T_{\text{axis}} \cdot \left[J_0(\kappa_0 R)\right]^{1/4}$$
5. **Self-consistent albedo (Rosen formula):**
$$\alpha_w = 1 - \frac{\dot{E}_w(T_{\text{int}},\, t_{\text{exp}})}{\sigma_{\text{SB}}\, T^4_{\text{int}}}$$